

Multi-scale Conformer Fusion Network for Multi-participant Behavior Analysis

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ABSTRACT

Understanding and elucidating human behavior across diverse scenarios represents a pivotal research challenge in pursuing seamless human-computer interaction. However, previous research on multi-participant dialogues has mostly relied on proprietary datasets, which are not standardized and openly accessible. To propel advancements in this domain, the MultiMediate'23 Challenge presents two sub-challenges: Eye contact detection and Next speaker prediction, aiming to foster a comprehensive understanding of multi-participant behavior. To tackle these challenges, we propose a multi-scale conformer fusion network (MSCFN) for enhancing the perception of multi-participant group behaviors. The conformer block combines the strengths of transformers and convolution networks to facilitate the establishment of global and local contextual relationships between sequences. Then the output features from all Conformer blocks are concatenated to fusion multi-scale representations. Our proposed method was evaluated using the officially provided dataset, and it achieves the best and second best performance in next speaker prediction and gaze detection tasks of MultiMediate'23, respectively.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence.

KEYWORDS

Eye contact detection; Next speaker prediction; Multi-scale conformer; Multi-participant behavior analysis

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1 INTRODUCTION

Continuous efforts are being made to propel technological advancements that enable machines to interact more naturally with humans. Central to this endeavor is the ability to perceive and interpret human behavior across various scenarios and tasks. In multi-party conversations, individual personalities play a role in their observable behavior. Yet, changes in human behavior and diverse interaction styles can lead to altered outcomes, resulting in valuable ideas being disregarded. Therefore, effectively analyzing human behavior in multi-party conversations has emerged as a challenging and crucial research topic. To address this challenge, a prevalent approach is to enhance interaction outcomes by supporting human dialogues through artificial mediators using the Wizard-of-Oz paradigm. However, group behavior is often diverse, and the rigid rule-based methods [24] may not adequately adapt to new or different situations, leading to sub-optimal results. In daily life, human interactions involve more than just verbal communication (e.g. speech) through words. Nonverbal behavioral cues, including eye contact, body posture, and facial expressions, play an integral role in interpersonal exchanges. To facilitate group behavior perception and analysis, the MultiMediate challenge introduces two explicit sub-challenges: *Eye contact detection* and *Next speaker prediction*. These tasks aim to improve understanding and enable seamless interaction with human groups, providing valuable insights into group dynamics [3, 18–20].

1.1 Eye Contact Detection

Eye contact is a vital non-verbal communication method in interpersonal interactions, providing valuable information about attention, emotions, and intentions [13, 23]. Eye contact detection has emerged as a useful tool for simulating human social skills and enhancing human-computer interaction [14, 22]. Eye contact detection refers to the automated identification of whether a person's gaze is directed towards another participant's eyes or face, as well as other objects of interest, etc [5]. In eye contact detection, a common approach involves estimating the gaze direction of each participant directly from the image [15, 26]. Previous methods often rely on extracting features such as the cornea and eye center from the facial

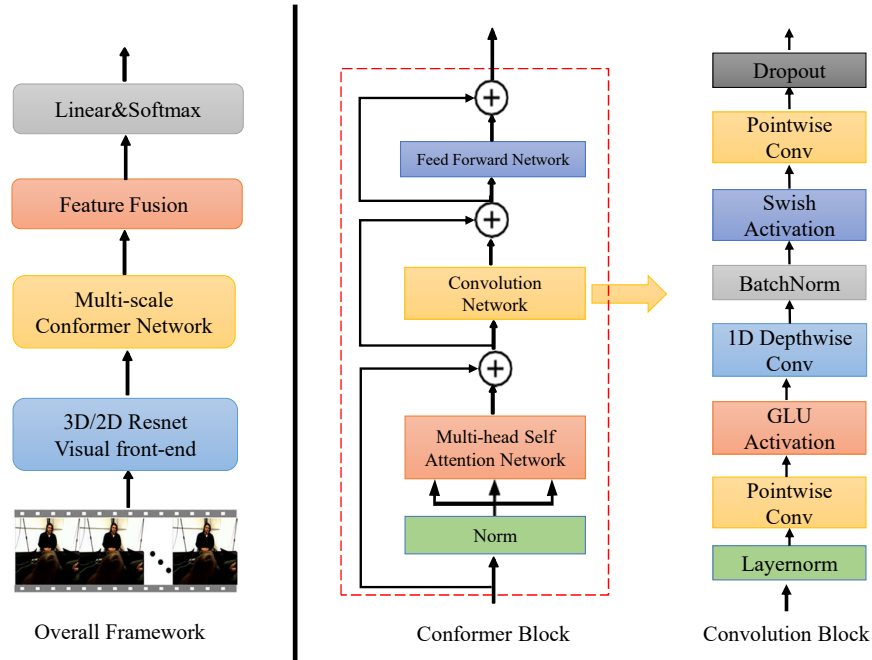


Figure 1: Overview of the proposed multi-scale conformer fusion network (MSCFN) for multi-participant behavior analysis. The conformer block is an important component of our framework.

region. Still, they primarily focus on a single-eye target and the participant closest to the camera. This approach faces challenges when multiple individuals are engaged in a conversation, as accurately extracting eye features becomes difficult due to varying distances between the participants and the camera. Recently, the correlation between gaze and speaking behaviors [22] was utilized to address the challenge of specific annotations for participant seat positions. Zhang et al. [31] employ OpenFace for eye gaze and head pose estimation and utilized a multi-layer perceptron for deep feature extraction. Due to the successful application of convolutional neural network (CNN) in video and image, Fu et al. [8] introduce a basic CNN model that jointly models participants' appearance and behavioral characteristics to detect eye contact.

1.2 Next Speaker Prediction

Next speaker prediction involves analyzing contextual information in a dialogue to understand and predict the transition of participant roles in the conversation [6, 25]. It enhances the interactive capability of dialogue systems, making them more intelligent and natural. In multi-party dialogues, accurately comprehending the speaking order of participants and avoiding speech conflicts is crucial for seamless participation in specific situations. To simulate and understand the skill of next speaker prediction, researchers employ machine learning methods that combine features like head movement [10], respiration [11], and blinking [7]. However, this approach presents a challenge for the next speaker prediction as each prediction may yield multiple candidates. Skantze et al. [28] propose a continuous turn-by-turn model that integrates acoustic

and part-of-speech features for speech activity prediction, employing a multimodal LSTM network. Building upon this approach, Roddy et al. [27] extended the methodology using a multi-scale recurrent neural network (RNN) architecture, utilizing separate LSTM sub-networks to model each modality. Birmingham et al. [4] combine group-level visual attention focus features and audio-video synchronizer models to improve next speaker prediction. Furthermore, to address the challenge of next speaker prediction in the presence of multiple candidates, a novel gated multi-modal fusion mechanism [30] has been designed to integrate various types of information for improving turn-by-turn prediction accuracy.

Although progress has been made in multi-participant behavior analysis, it remains a challenging task. The introduction of the MPIIGroupInteraction corpus has provided a valuable resource for research in this domain. Previous approaches typically involved using OpenFace for feature extraction, employing time series networks for modeling, or utilizing more experimental data for group behavior analysis. In contrast, we propose an end-to-end model that directly operates on the original video sequence. The multi-scale conformer fusion framework is proposed to model these features contextually. This approach effectively captures the local and global relationships within the input sequence features, offering a promising solution for multi-speaker behavior analysis.

2 METHOD

In this section, We describe the implementation of the proposed multi-scale conformer fusion network (MSCFN) for these two challenges in detail. Then the task of Eye contact detection and Next

speaker prediction is defined. The proposed framework overview is described in Figure 1.

2.1 Visual Front-end

We first obtain vision sequence frames from the raw video at a sampling rate of 25 frames per second. The frames are resized to 112x112 pixels and converted to grayscale. We employ a 2D ResNet-18 architecture to extract visual features, processing each frame independently. The output is a 521-dimensional vector representing the visual features. This visual front-end feature extraction allows us to capture relevant visual information for Eye contact detection and Next speaker prediction tasks. The architecture details can be found in Table 1.

Table 1: Visual Front-end architecture.

Stage	Layers	Output Shape
Stem	Conv3d: $5 \times 7^2, 64\text{filters}$ MaxPoo3d: 1×3^2	$(64, T_v, 56, 56)$
Residual Block ₁	Conv 2D, $3^2, 64$ Conv 2D, $3^2, 64$ $\times 2$	$(64, T_v, 28, 28)$
Residual Block ₂	Conv 2D, $3^2, 128$ Conv 2D, $3^2, 128$ $\times 2$	$(128, T_v, 14, 14)$
Residual Block ₃	Conv 2D, $3^2, 256$ Conv 2D, $3^2, 256$ $\times 2$	$(256, T_v, 7, 7)$
Residual Block ₄	Conv 2D, $3^2, 512$ Conv 2D, $3^2, 512$ $\times 2$	$(512, T_v, 4, 4)$
Pool	Global Average Pooling	$(T_v, 512, 1, 1)$
Reshape	–	$(T_v, 512)$

2.2 Multi-scale Conformer Fusion Network

Transformer architecture [29] is known for its outstanding capabilities in long-range sequence modeling, making it highly suitable for a range of tasks [1, 2, 16, 17]. In contrast, Transformer can be less effective at modeling local information. To address this limitation and leverage both global and local information, the Conformer framework has been proposed to integrate the strengths of both Transformers and convolutional modules. This integration allows Conformer models to effectively model both global and local contextual relationships within the features. Therefore, we introduce the Conformer block [9] to capture better and analyze the multi-participant behavior.

As shown in Figure 1, the conformer architecture is mainly composed of multi-head self-attention (MHSA) and convolution components. The convolution module after MHSA in the Conformer architecture includes Pointwise convolution, 1D Depthwise convolution, and BatchNorm. Mathematically, the input features z_{i-1} to the i Conformer block, and the output features z_i are obtained.

The calculation is as follows:

$$\tilde{z}_i = z_{i-1} + \frac{1}{2} \text{FFN}(z_{i-1}) \quad (1)$$

$$z'_i = \tilde{z}_i + \text{MHSA}(\tilde{z}_i) \quad (2)$$

$$z''_i = z'_i + \text{Conv}(z'_i) \quad (3)$$

$$z_i = \text{LayerNorm}(z''_i + \frac{1}{2} \text{FFN}(z''_i)) \quad (4)$$

Here, $\text{FFN}(\cdot)$, $\text{MHSA}(\cdot)$, $\text{Conv}(\cdot)$ and $\text{LayerNorm}(\cdot)$ stand for a feed-forward network, a multi-head self-attention, a convolution and a layer normalization operation respectively. $\{z_{i-1}, z_i\} \in \mathbb{R}^{d \times T}$, where d denotes the Conformer encoder dimension and T denotes the length of input sequence.

Furthermore, to effectively leverage the features from different levels, we aggregate the feature maps as shown in Figure 2. This involves combining the feature maps from all Conformer modules into a single representation. The aggregation can be done by concatenating the feature maps along the channel dimension:

$$Z = \text{Concat}(z_1, z_2, \dots, z_N) \quad (5)$$

where $Z \in \mathbb{R}^{D \times T}$ and N denotes the number of Conformer blocks.

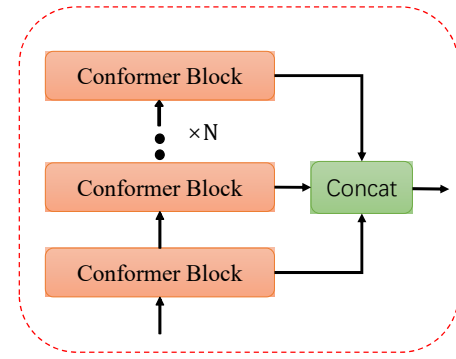


Figure 2: Overview of multi-scale conformer feature fusion.

2.3 Task Definition

Eye contact was defined as a discrete indication of which participant a certain participant was making eye contact with. Each instance in the dataset has 10 seconds of audio-visual data, providing contextual information for detecting eye contact. We define eye contact detection as a multi-class classification task for inferring each participant's location and eye contact status. The next-speaker prediction task refers to predicting a participant who will speak at a fixed point in the future in a context window containing multiple potential speakers. Unlike the gaze detection task, this is a multi-label classification task. The goal is to predict the state of each potential speaker (i.e. speaking or not speaking).

3 EXPERIMENTS AND RESULTS

3.1 Dataset

By leveraging the MPIIGroupInteraction dataset [21], we validate the effectiveness and robustness of our approach in tasks such as

Eye contact detection and Next speaker prediction within multi-party conversations. This dataset comprises 22 group conversations between 3 to 4 participants, with each conversation spanning a duration of 20 minutes. The dataset provides subsets of different sizes for training, validation, and testing.

3.2 Implementation

Our method is implemented using the PyTorch framework. The computational platform for training and evaluation consists of an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. The conformer encoder dimensions and the number of MSHA heads are set to 512 and 8, respectively. The number of conformer block N is assigned to [1, 2, 4, 6]. The backbone ResNet-18 is initialized with ImageNet pretrained weights. We employ the Adam optimizer [12] with hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.98$, and $\epsilon = 10^{-9}$. The initial learning rate is set to 0.0001. We employ a linear warm-up followed by a cosine-decay learning rate schedule to update the learning rate. During the evaluation, we use the average model over the last five checkpoints to obtain reliable performance metrics. To prevent overfitting, we apply data augmentation techniques, such as random cropping and horizontal flipping, to the image frames. Eye contact detection is evaluated using accuracy, while next speaker prediction is evaluated using unweighted average recall (UAR). Our approach adopts an end-to-end training strategy for the two sub-tasks of eye contact detection and next speaker prediction. Unlike previous methods [4], we do not rely on external tools like open face for feature extraction or pre-processing.

Table 2: Accuracies (ACC) performance of our method and comparison methods. N denotes the number of transformer/conformer blocks.

Methods	ACC(val)	ACC(test)
Baseline[19]	0.54	0.52
Baseline[3]	0.64	0.57
UME[8]	-	0.56
MSTFN ($N=1$)	0.56	-
MSCFN ($N=1$)	0.60	-
MSCFN ($N=2$)	0.63	0.59
MSCFN ($N=4$)	0.68	0.60
MSCFN ($N=6$)	0.72	0.65

3.3 Result

Result of Eye Contact Detection task: Our method operates directly on raw video data. The official baseline method employs OpenFace for gaze and head poses feature extraction, followed by fusion using RBF-SVM. The multi-scale transformer fusion network (MSTFN) is obtained by removing the Convolution Block in MSCFN. From the results in the Table 2, the performance of MSCFN is comparable to that of MSTFN when $N=1$. This indicates that the conformer has a better feature representation ability compared to the transformer. To investigate the impact of different scales, we experiment with different numbers of Conformer blocks, denoted as $N = [1, 2, 4, 6]$. As N increases to 6, our method outperforms the

Table 3: Unweighted Average Recall (UAR) performance of our method and comparison methods. N denotes the number of transformer/conformer blocks.

Methods	UAR(val)	UAR(test)
Baseline[19]	0.60	0.51
Anonymous ¹	-	0.53
JYNB ²	-	0.55
MSTFN ($N=1$)	0.52	-
MSCFN ($N=1$)	0.58	-
MSCFN ($N=2$)	0.61	-
MSCFN ($N=4$)	0.65	0.52
MSCFN ($N=6$)	0.70	0.58

baseline by 0.08, achieving the best accuracy of 0.65. The results demonstrate that incorporating feature clustering at different scales can indeed improve the performance of the model. By considering multiple scales, the model can capture and leverage information at different levels of granularity, leading to enhanced representation and more accurate predictions.

Result of Next Speaker Prediction task: We also explore the impact of the number of conformer blocks in the task of next speaker prediction. The experimental comparison results of our proposed method with several existing methods are reported in Table 3. Notably, when N is set to 6, our method achieves the highest performance with a UAR of 0.58 among the compared methods. This indicates that our approach effectively captures the behavioral characteristics of multi-participant in the conversation.

4 CONCLUSION

We introduce a multi-scale conformer fusion network for the Eye Contact Detection and Next Speaker Prediction sub-challenges of MultiMediate'23, aiming to perceive and analyze the behavior of multiple participants accurately. Our method operates directly on raw video data in an end-to-end manner. The experimental results on the validation set and the unpublished test set demonstrate our approach's significant superiority and potential. Notably, our method achieves the second-best and best performance in the final test results of MultiMediate'23.

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